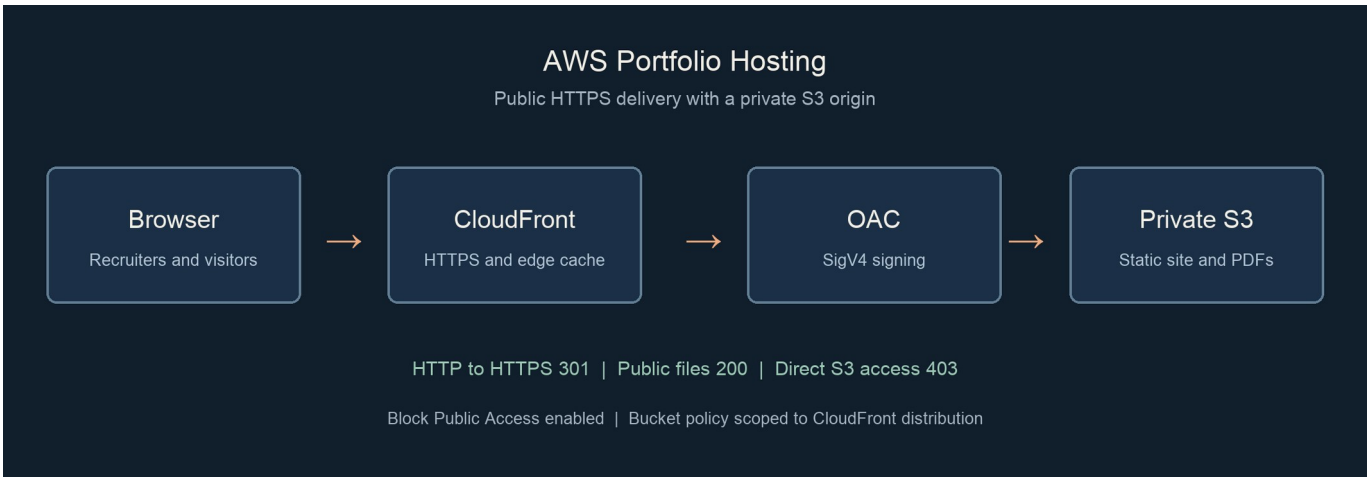
Secure AWS Portfolio Hosting

Live personal website | Terraform and GitHub Actions | October 2026

Public portfolio

Business goal and solution

Make my engineering work, resume and sanitized project evidence accessible to recruiters through a public HTTPS website. Managed existing private S3/CloudFront hosting with Terraform and automated content releases using GitHub Actions and OIDC, without operating a server.



Visitors reach CloudFront; OAC signs requests to the private S3 origin. Direct unauthenticated S3 access is denied.

Implementation and security

- Imported five existing hosting resources into Terraform without replacement; preserved private S3, SigV4 OAC and the existing WAF association. Added separate encrypted, versioned state with native S3 locking.
- Implemented validation and automatic main-branch delivery with pinned GitHub Actions and short-lived OIDC credentials. The role can write site files and invalidate this distribution; it cannot modify infrastructure or read state.
- Uploaded immutable assets before promoting the root entry page; invalidated root cache paths and verified public SHA-256 checksums. A manual workflow restores a saved release after validating its manifest.

Verified results

Eight release tests passed. Live releases A and B each verified all 12 public files; rollback from B to saved release A succeeded. The refreshed Terraform plan showed no changes. Public HTTPS returned 200, HTTP redirected to HTTPS, and direct S3 access returned 403. Desktop and mobile checks passed.

Source and recorded deployment and rollback runs

Lessons and scope

Publish the entry page last, retain immutable assets and verify actual rollback. Infrastructure applies remain operator-reviewed; website deployment is automatic. No custom domain, Route 53, load test or uptime SLA is claimed. State and release storage, requests and delivery remain subject to AWS pricing.